

AI & Machine Learning Internship – Task 4

Classification Models, Evaluation Metrics & Handling Imbalanced Data

Objective

The objective of this project was to build and evaluate a binary classification model using the Breast Cancer Dataset. The project focused on understanding classification metrics, evaluating model performance beyond accuracy, handling class imbalance, and comparing multiple classifiers.

Dataset Description

The Breast Cancer Dataset from Scikit-learn was used in this project. The dataset contains features extracted from breast cancer cell images and is used to classify tumors as malignant or benign.

Target Classes:

- 0 = Malignant
- 1 = Benign

The dataset was split into training and testing sets using an 80:20 ratio with stratification to preserve class distribution.

Data Preprocessing

Feature scaling was performed using StandardScaler to normalize feature values before training the Logistic Regression model.

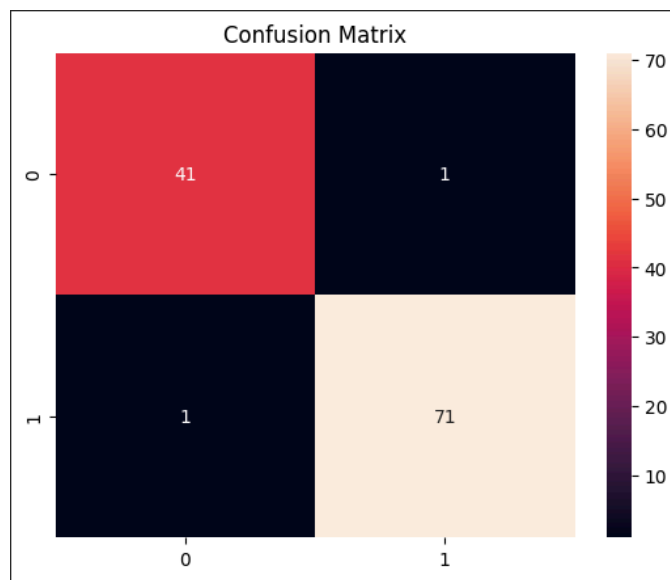
The dataset was divided into training and testing datasets using `train_test_split()` with stratification to ensure balanced class representation.

Logistic Regression Model

A Logistic Regression classifier was trained as the baseline classification model.

Confusion Matrix

The confusion matrix obtained was:



This indicates:

- True Negatives (TN) = 41
- False Positives (FP) = 1
- False Negatives (FN) = 1
- True Positives (TP) = 71

The model correctly classified most instances with very few errors.

Confusion Matrix Interpretation

True Positives (TP)

True Positives are cases where the model correctly predicts a positive class. In this project, 71 benign cases were correctly classified.

True Negatives (TN)

True Negatives are cases where the model correctly predicts a negative class. In this project, 41 malignant cases were correctly classified.

False Positives (FP)

False Positives occur when the model incorrectly predicts a positive class when the actual class is negative. In this project, only 1 case was falsely classified as positive.

False Negatives (FN)

False Negatives occur when the model incorrectly predicts a negative class when the actual class is positive. In this project, only 1 case was falsely classified as negative.

Classification Metrics

The Logistic Regression model achieved the following results:

- Accuracy = 98.25%
- Precision = 0.99
- Recall = 0.99
- F1-Score = 0.99

The results indicate excellent classification performance with very few prediction errors.

	precision	recall	f1-score	support
0	0.98	0.98	0.98	42
1	0.99	0.99	0.99	72
accuracy			0.98	114
macro avg	0.98	0.98	0.98	114
weighted avg	0.98	0.98	0.98	114

Questions and Answers

Why is accuracy alone insufficient?

Accuracy can be misleading when datasets are imbalanced. A model may achieve high accuracy by predicting the majority class most of the time while failing to correctly identify important minority class instances. Therefore, Precision, Recall, and F1-Score should also be considered.

Which metric is more important for medical diagnosis?

Recall is generally more important in medical diagnosis because it measures how many actual positive cases are correctly identified. Missing a disease case can have serious consequences for patients.

What happens if recall is low?

A low recall value means the model is failing to identify many actual positive cases. This increases the number of False Negatives, which can be dangerous in applications such as disease detection and fraud detection.

Why is F1-Score preferred for imbalanced data?

F1-Score combines Precision and Recall into a single metric. It provides a balanced evaluation of model performance and is especially useful when class distributions are unequal.

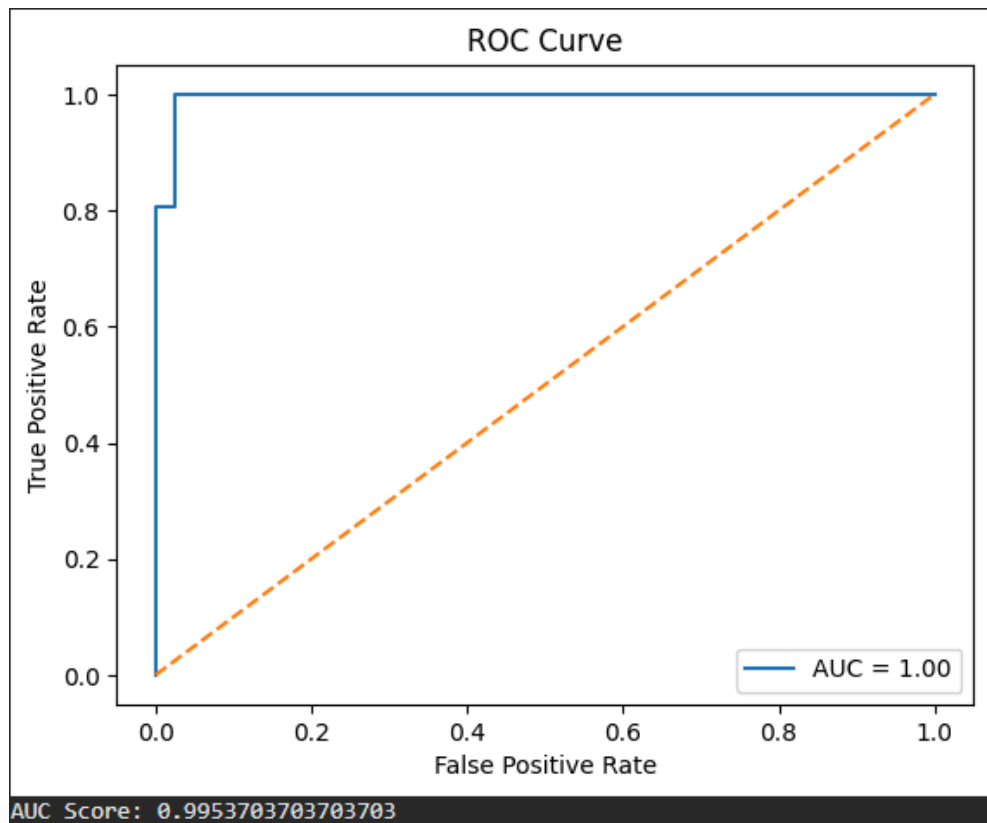
ROC Curve and AUC Score

The ROC Curve was generated to evaluate the classifier's ability to distinguish between classes.

AUC Score:

0.9954

An AUC value close to 1 indicates excellent classification performance and strong class separation. The obtained score demonstrates that the model can effectively distinguish between malignant and benign cases.



Handling Class Imbalance

Class imbalance was addressed using Logistic Regression with:

```
class_weight = "balanced"
```

This approach automatically adjusts class weights according to class frequencies and helps the model pay equal attention to both classes.

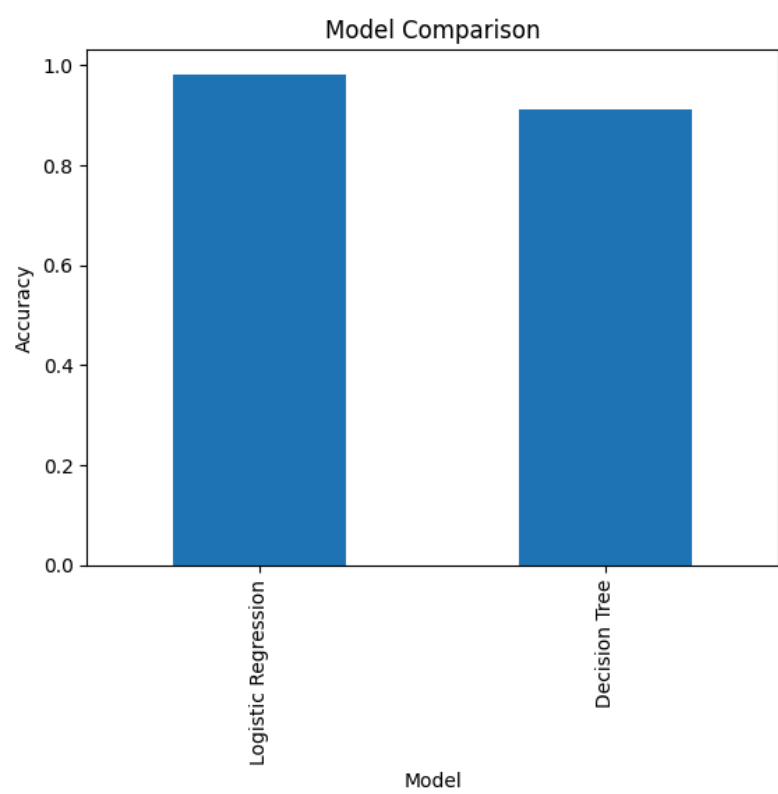
Using class weights is a commonly used technique in real-world machine learning applications where one class may have significantly fewer examples than another.

Model Comparison

Two classification models were compared:

Model	Accuracy
Logistic Regression	98.25%
Decision Tree	91.23%

The Logistic Regression model achieved higher accuracy and better overall performance.



Logistic Regression vs Decision Tree

Logistic Regression achieved:

- Higher Accuracy
- Higher Precision
- Higher Recall
- Higher F1-Score
- Better Generalization

The Decision Tree classifier showed lower performance and is generally more susceptible to overfitting.

Therefore, Logistic Regression was selected as the preferred model for this classification task.

Final Model Decision

Logistic Regression was selected as the final model because it achieved:

- Accuracy = 98.25%
- Precision = 0.99
- Recall = 0.99
- F1-Score = 0.99
- AUC Score = 0.9954

These results demonstrate excellent predictive performance and strong classification capability.

Conclusion

This project demonstrated the complete workflow of a classification problem, including data preprocessing, feature scaling, model training, confusion matrix analysis, evaluation metrics, ROC-AUC analysis, class imbalance handling, and model comparison.

Among the tested models, Logistic Regression achieved the best overall performance with an accuracy of 98.25% and an AUC score of 0.9954, making it the most suitable model for this classification task.

The project provided practical experience in evaluating classification models using multiple metrics and selecting models based on scientific performance analysis rather than accuracy alone.